



**PES University, Bengaluru**  
(Established under Karnataka Act 16 of 2013)

**END SEMESTER ASSESSMENT (ESA) - Dec 2024**

**UE21CS352B - Object Oriented Analysis & Design using Java**

**Total Marks : 100.0**

1.a. Draw a use case diagram for railway reservation system which has all basic functionalities similar to IRCTC portal by identifying at least 5 usecases. (6.0 Marks)

1.b. With respect to class diagram explain the following terms with an example:  
Composition  
Aggregation  
Association  
Qualified Association  
Generalization  
Recursive relation (12.0 Marks)

1.c. For the below given scenario draw an activity diagram

In a Flight Reservation System, a customer books a flight through an online platform. The booking process involves several subsystems and conditional flows like checking for flights, selecting seats, processing payment, and notifying customers.

The customer searches for available flights based on origin, destination, and date. The customer enters personal details, including name, contact info, and travel document details to book a ticket. The customer decides whether to add services like extra baggage, in-flight meals, or travel insurance. This step is optional. A confirmation email or SMS with the booking details is sent to the customer. System should also handle payment failures. (7.0 Marks)

2.a. Design a class diagram for a system to manage online shopping based on the information given below. In an online shopping system, several complex subsystems need to work together to place an order. These subsystems can include:

1. Inventory System – Checks if the items are in stock. (Handle different types of items like perishable and non-perishable)
2. Payment System – Processes the payment.
3. Shipping System – Handles shipping details.
4. Notification System – Sends order confirmation emails to the customer.

Ensure that the constructors and destructors are appropriately used for each class. Demonstrate overloading of methods to display details of both perishable and non perishable items. (7.0 Marks)

2.b. Answer the following:

1. What is the need for interfaces in java.
2. Write a Java program that demonstrates the use of an abstract class. The program should include the following:
  - An abstract class Shape with an abstract method area() to calculate the area of a shape.
  - Two subclasses Circle and Rectangle that extend Shape. Each subclass should implement the area() method to calculate the area of a circle and a rectangle respectively.
  - In the main method, create objects of both Circle and Rectangle and calculate the area for both shapes.

(6.0 Marks)

2.c. Write a Java program to demonstrate the use of the super keyword in the following scenarios:

1. Accessing a superclass constructor from a subclass.
2. Accessing a superclass method from a subclass.

(6.0 Marks)

2.d. Answer the following

i) What is the output of the below given program?

```
public class Example {  
    static int count = 0;  
    public Example() {  
count++;  
    }  
    public static void displayCount() {  
        System.out.println("Count: " + count);  
    }  
    public static void main(String[] args) {  
        Example obj1 = new Example();  
        Example obj2 = new Example();  
        Example.displayCount();  
    }  
}}
```

ii) State the difference between static and non-static members in Java.

(6.0 Marks)

3.a. A library management system needs to keep track of books borrowed by members. Each member can borrow multiple books, and the system must calculate the total number of books borrowed by all members in the library. Using the Information Expert principle, identify which class should be responsible for calculating the total number of borrowed books. Justify your choice and design the solution.

(6.0 Marks)

3.b. Suppose you have a `Payment_Processor` class that handles credit card payments and sends email notifications.

Which SOLID principles does it violate and why?

How would you redesign it?

(6.0 Marks)

3.c. Explain lazy and early implementation of Singleton Design pattern using Java. Highlight its advantages and disadvantages

(6.0 Marks)

3.d. Explain the builder pattern with one relevant example

(7.0 Marks)



4.a. Differentiate between pattern and anti-pattern

(6.0 Marks)

4.b. What is a proxy pattern? Which are the three main variations of proxy pattern. (6.0 Marks)

4.c. What are anti patterns? State the solution for Blob, Lava Flow, Vendor lockin and Analysis-Paralysis antipatterns. (5.0 Marks)

4.d. Explain the Facade Design Pattern with a suitable real-world example. Discuss how it helps simplify client interaction with complex systems. Provide a brief code example to illustrate your explanation. (8.0 Marks)

**END SEMESTER ASSESSMENT (ESA) - May 2023****UE20CS352 - Object Oriented Analysis & Design with Java****Total Marks : 100.0**

1.a. Draw the Class Diagram to model a system for management of courses and enrolments. Identify classes, attributes, relationships and cardinality.

1. A Department manages multiple Sessions.
2. A Session has a Start date, an End date and Name (like Jan-May 2023).
3. A session can have multiple semesters running (like 6<sup>th</sup>, 4<sup>th</sup>, 2<sup>nd</sup> etc).
4. A student belongs to a semester and to a section within it. A section has minimum of 60 students.
5. A semester has a set of courses offered in it.
6. A student has to enrol for all the required number of courses (let us assume 5 for the listed semesters). For some courses like electives, a minimum of 20 students have to enrol for it to be offered.
7. An enrolled student's attendance for the course is recorded.
8. Similarly, an enrolled student's ISA, ESA marks and Grade for the course are recorded.
9. A teacher will be assigned to a section for a course.
10. A teacher can also be assigned as the class teacher for a particular section

Note: Assume attributes and other details as appropriate.

(6.0 Marks)

1.b. For the given scenarios, choose appropriate diagram to model the scenario. Justify your choice and draw the chosen diagram for the scenario.

- i) Every student is assigned a Faculty Advisor. The Faculty Advisor is assigned a minimum of 25 students with no maximum limit.
- ii) An Order consists of multiple Items. The Order total is computed by adding the product of Quantity and Price for each Item in the Order.

(6.0 Marks)

1.c. Write short notes about the following features of a **State diagram** with appropriate notations and an example.

- i) Order of activities associated with a state transition
- ii) Synchronization and Splitting associated with concurrent states (4.0 Marks)

1.d. Write about the elements in a Component Diagram. Represent the same in an example diagram using the correct notations.

(4.0 Marks)

2.a. What are the three types of constructors that can be defined in Java? Give an example of each. (6.0 Marks)

2.b. Using method overloading in Java, write a class NextValue with a method getNextValue that takes either integer or float as input and returns next integer in case the parameter is integer or returns a float value by adding 0.01 in case of float input.

Write the main method that calls the method with different parameters.(4.0 Marks)

2.c. Write Java code for the given scenario.

1. The SHIELD assigns the Avengers to a Mission.
2. The Mission has a name, location, strength requirement and list of Avengers assigned.
3. An Avenger has name, weapon and strength level.
4. The addAvenger method in Mission, takes a list of all Avengers, iterates through the list and assigns all those Avengers to the Mission who has a strength level greater than the Mission's strength requirement.
5. The showMission method should display the Mission details and list of Avengers assigned using System.out.println(avenger) directly.

The main method is given below for your reference with the Expected Output.

```
public class MissionMain {  
    public static void main(String[] args) {  
        Avengers ironMan = new Avengers("Iron Man", "Suit", 9);  
        Avengers thor = new Avengers("Thor", "Hammer", 10);  
        Avengers captainAmerica = new Avengers("Captain America", "Shield", 8);  
        Avengers blackWidow = new Avengers("Black Widow", "Guns", 7);  
  
        List<Avengers> avengersList = new ArrayList<>();  
        avengersList.add(ironMan);  
        avengersList.add(thor);  
        avengersList.add(captainAmerica);  
        avengersList.add(blackWidow);  
  
        Mission mission = new Mission("Save the World", "New York", 9);  
        mission.addAvenger(avengersList);  
        mission.showMission();  
    }  
}
```

**Expected Output:**

Save the World New York 9  
Iron Man Suit 9  
Thor Hammer 10

(8.0 Marks)

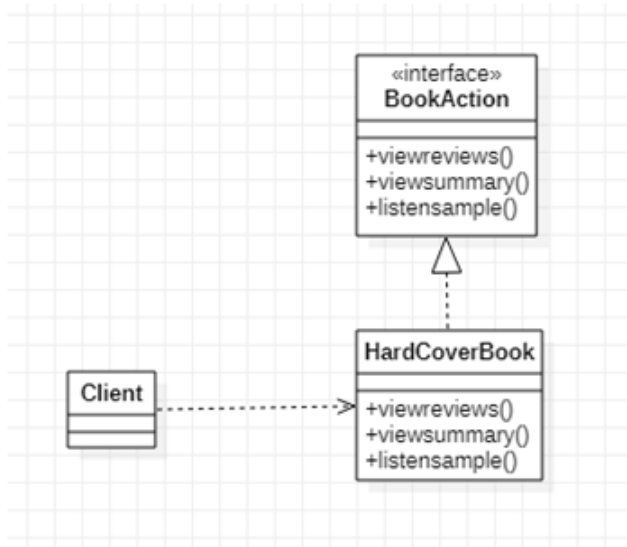
2.d. Is the below statement valid in Java? Justify your answer.

Object o = new ArrayList<String>();

(2.0 Marks)

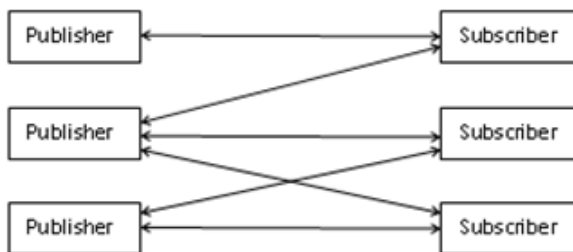
3.a. Identify the three SOLID principles violated by this design. Suggest correction so that the violated principles are adhered to.

(6.0 Marks)



3.b. What is wrong with the model given below in terms of GRASP design principles? Show how can this be improved with a structural diagram.

(6.0 Marks)



3.c. With a neat flow diagram, explain the working of MVC architecture pattern.

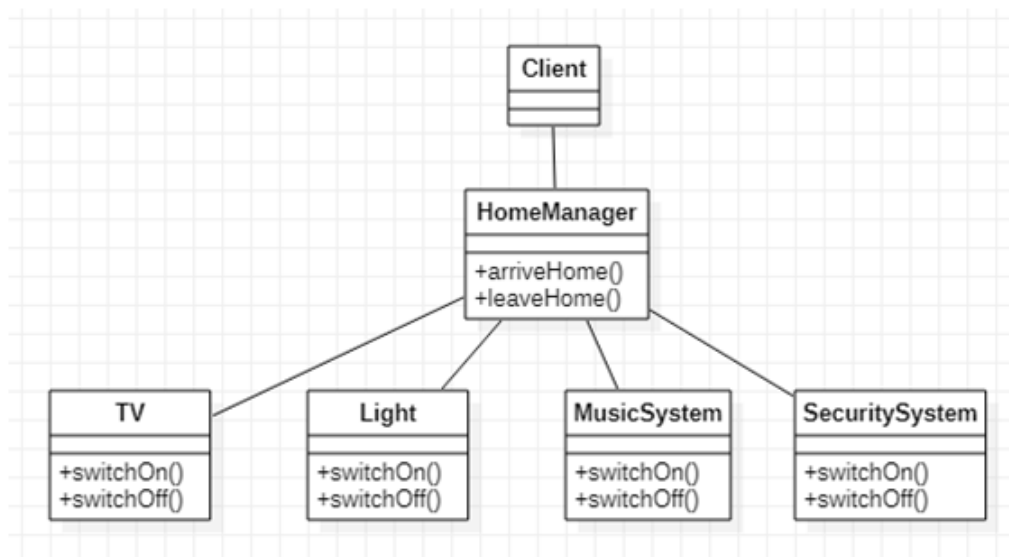
(4.0 Marks)

3.d. Explain the Liskov Substitution Principle using an example where LSP is violated and how it can be corrected. (4.0 Marks)

4.a. Describe the Factory Method creational pattern with a suitable example design and corresponding code snippet. (6.0 Marks)

4.b. Identify the structural pattern being used in the design diagram below. Justify. Write code to implement the HomeManager class constructor and methods.

(6.0 Marks)



4.c. Implement the following use case using a proxy pattern. Write both the class diagram and the code snippet. State which type of proxy pattern is suitable for the given use case.

Use Case:

If an entire library of songs, with all the details about each song, is loaded from a database, it will consume a lot of RAM, and it is very likely that the user will need to play *only* one song. The solution displays only the *title*, *artist*, and *duration* of the songs when a list of them is displayed. When a song is selected, the corresponding media stream will be fetched from the database and operations like play can be performed.

Note: do not write the entire code; write only code snippets/pseudo code to instantiate and use the proxy and the real object. (6.0 Marks)

4.d. Fill in the blanks.

- i) \_\_\_\_\_ pattern refers to creating duplicate object while keeping performance in mind.
- ii) \_\_\_\_\_ pattern provides the interface according to client requirement while using the services of a class with a different interface (2.0 Marks)

5.a. Describe the structure of the solution for **Iterator** pattern. (4.0 Marks)



5.b. Implement the following use case using a Chain of Responsibility pattern. Write both the class diagram and the code snippet for building the chain.

A logging system provides functionality to write INFO messages to the console (terminal), ERROR messages to an error log file and DEBUG messages to a debug log file. The client invokes the console logger irrespective of the type of message. The console logger will check the type and if the message is INFO message then it writes it to the console. Else, it invokes the next level logger that is the error logger. The error logger does the same and if the message is not an ERROR message it invokes the debug logger. The same applies to DEBUG logger, however, it does not have any next level logger. (6.0 Marks)

5.c. Explain with an example the steps for the refactored solution for the Blob Anti-pattern. (6.0 Marks)

5.d. Mention any two symptoms and any two typical causes for Analysis Paralysis design anti-pattern. (4.0 Marks)

**END SEMESTER ASSESSMENT (ESA) - JULY - 2023****UE20CS352 - Object Oriented Analysis & Design with Java****Total Marks : 100.0**

1.a. Write briefly, with appropriate examples, about the Analysis and Design phases in Object Oriented Approach for software development. Distinguish this approach from the Procedure oriented approach. (6.0 Marks)

1.b. Identify the use cases and actors and their relationships for Passport Automation System and depict it with a Use Case diagram:

**Passport Automation System:** To simplify the process of applying for a passport, a software system has to be created. Initially the applicant will login to the passport automation system and submit his details. These details are stored in the database and verification process is done by the passport administrator, the regional administrator and the police, then the passport is issued to the applicant.

- Passport Automation System is used in the effective dispatch of passport to all of the applicants. This system adopts a comprehensive approach to minimize the manual work and schedule resources, time in a cogent manner.

- The core of the system is to get the online registration form (with details such as name, address etc.,) filled by the applicant whose testament is verified for its genuineness by the Passport Automation System with respect to the already existing information in the database.

- This forms the first and foremost step in the processing of passport application. After the first round of verification done by the system, the information is in turn forwarded to the regional administrator's (Ministry of External Affairs) office.

- The applicant can optionally seek ECNR status enabled by providing Graduation Certificates and Employment details. This is processed by the Passport Administrator.

- The application is then processed manually based on the report given by the system, and any forfeiting identified can make the applicant liable to penalty as per the law.

- The system forwards the necessary details to the police for its separate verification whose report is then presented to the administrator. After all the necessary criteria have been met, the original information is added to the database and the passport is sent to the applicant. (8.0 Marks)

1.c. What are the different elements in a Deployment Diagram? Write the notations for these in an example diagram. (6.0 Marks)

2.a. Write about the three access specifiers i.e. private, protected and public, that can be used in Java. Using one example class, briefly write about the effect of using the access specifiers on member variables. (6.0 Marks)

2.b. Write code to add the following functionality expected by the MainClass. The Employee object can be created in two ways. The Manager is an Employee. In addition, the Manager has an array/list of Employees who are his/her team members.

```
class Employee{  
    private String name;  
    private String dept;
```

```
    //Write code here  
}
```

```
// Write code here
```

```
class MainClass{  
    //Do not change this method  
    public static void main(String[] args) {  
        Employee e1 = new Employee("Employee Name");  
        Employee e2 = new Employee ("Employee Name", " Employee Department");  
  
        Manager m1 = new Manager("Manager Name", "Manager Department");  
        m1.addTeamMember(e1);  
        m1.addTeamMember(e2);  
  
        m1.showDetails(); // shows details of Manager as well as all his/her team members  
    }  
}
```

Note: only write the code expected. Do not repeat the given code. (6.0 Marks)

2.c. Write short notes about the following. Use code examples as applicable.

1. Method overloading (2 marks)
  2. Static and Instance members in a class (3 marks)
  3. Abstract class (3 marks)
- (8.0 Marks)

3.a. In the given class model, the responsibility to compute the total marks a Test is being administered for, should be assigned to which class? The assignment of method is based on which principle? Write Java code snippet for the same.



(6.0 Marks)

3.b. State the Interface Segregation Principle. Explain the the problem and solution, with an appropriate example. (8.0 Marks)

3.c. Describe briefly the Pure Fabrication GRASP Design Principle, with an example. (6.0 Marks)

4.a. Write briefly about the Adapter Structural Design Pattern. Clearly state the intent, problem , solution structure and example. (6.0 Marks)

4.b. Implement the following use case using a proxy pattern. Write both the class diagram and the code snippet. State which type of proxy pattern is suitable for the given use case.

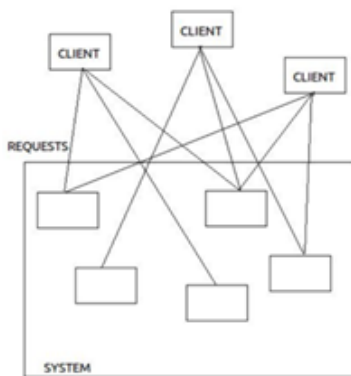
Use Case:

A department has thousands of students, with all their details Name, SRN, Dept, etc. At any given time, it is very likely that the user needs the grades of only one student. The solution displays only the *Name*, *SRN*, and *Dept* of the students when a list of them is displayed. When a student is selected, the grades of the corresponding student will be fetched from the database and displayed.

Note: do not write the entire code; write only code snippets/pseudo code to instantiate and use the proxy and the real object. (6.0 Marks)

4.c. What is the problem with the following system? What design pattern can be used to overcome that problem? Write the structure of the solution. State the advantages of using that solution.

(8.0 Marks)



5.a. Explain the Chain of Responsibilities behavioral pattern. State the intent, structure and benefits of this pattern. (6.0 Marks)

5.b. What is the refactored solution to tackle Vendor Lock-in Anti-pattern?  
(6.0 Marks)

5.c. Mention symptoms and typical causes for the Blob design anti-pattern.  
(8.0 Marks)



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**END SEMESTER ASSESSMENT (ESA) - DEC 2023**

**UE20CS352 - Object Oriented Analysis & Design with Java**

**Total Marks : 100.0**

1.a. State any four differences between Method overloading and Method overriding? (4.0 Marks)

1.b. State the reason why the given code prints output as false. Also change the code such that the output in the main function is true.

```
public class MainClass
{
    public static void main(String[] args)
    {
        Classdemo objOne=new Classdemo(10,20);
        Classdemo objTwo=new Classdemo(10,20);
        System.out.println(objOne.equals(objTwo)); //prints false
    }
}
class Classdemo
{
    int a; int b;
    Classdemo(int a,int b)
    {
        this.a=a;
        this.b=b;
    }
}
```

(5.0 Marks)



1.c. Elucidate need for an abstract class and justify the same with an example. (6.0 Marks)

1.d. **Give the output of java program given below:**

```
public class StaticDemo {  
    static int a=1,b;  
    static void display(int x){  
        System.out.println("x="+x);  
        System.out.println("a="+a);  
        System.out.println("b="+b);  
    }  
    static{  
        System.out.println("Inside static block");  
        b=a*2;  
    }  
    public static void main(String[] args) {  
        System.out.println("Inside main");  
        display(5);  
    }  
}
```

(5.0 Marks)

2.a. Elucidate qualified association and aggregation relationship w.r.t class diagram with an example for each. (6.0 Marks)

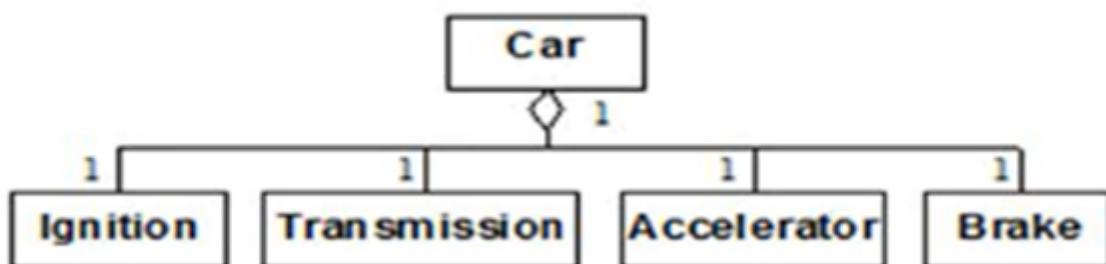
2.b. Draw a Use Case diagram of a Railway Reservation System. In the system, a Passenger logs to the system and checks for trains available. The passenger can book a ticket and optionally choose a seat of their choice and accordingly make a card payment for it. The bank validates the payment for the system. Passengers can also cancel their ticket. The system admin processes the cancellation as well as update train when required. (6.0 Marks)

2.c. Describe in a line or two the following elements of activity diagram with their UML notation:

- i) Activity
- ii) Object Flow

(4.0 Marks)

2.d. Draw a state diagram for Transmission component of a Car wrt the diagram given below



(4.0 Marks)

3.a. Explain the creator and controller GRASP principles with an example along with the corresponding class diagram (8.0 Marks)

3.b. Define MVC architectural pattern. Explain the working of MVC with a neat diagram (6.0 Marks)

3.c. Fill in the blanks:

- i. \_\_\_\_\_ Violating Dependency Inversion Principle will result in a code with \_\_\_\_\_ coupling.
- ii. \_\_\_\_\_ Liskov Substitution Principle states that the designer/implementer should be able to replace objects of \_\_\_\_\_ with the objects of \_\_\_\_\_.
- iii. \_\_\_\_\_ Principle states that the design should be extensible however modifying existing design should be avoided.
- iv. \_\_\_\_\_ The pattern that hides the complexity related to creation of objects from the client is \_\_\_\_\_.
- v. A \_\_\_\_\_ is a best practice, a recipe, a time tested solution to a recurring problem that's proved to work and to be the best one among to choose. (6.0 Marks)

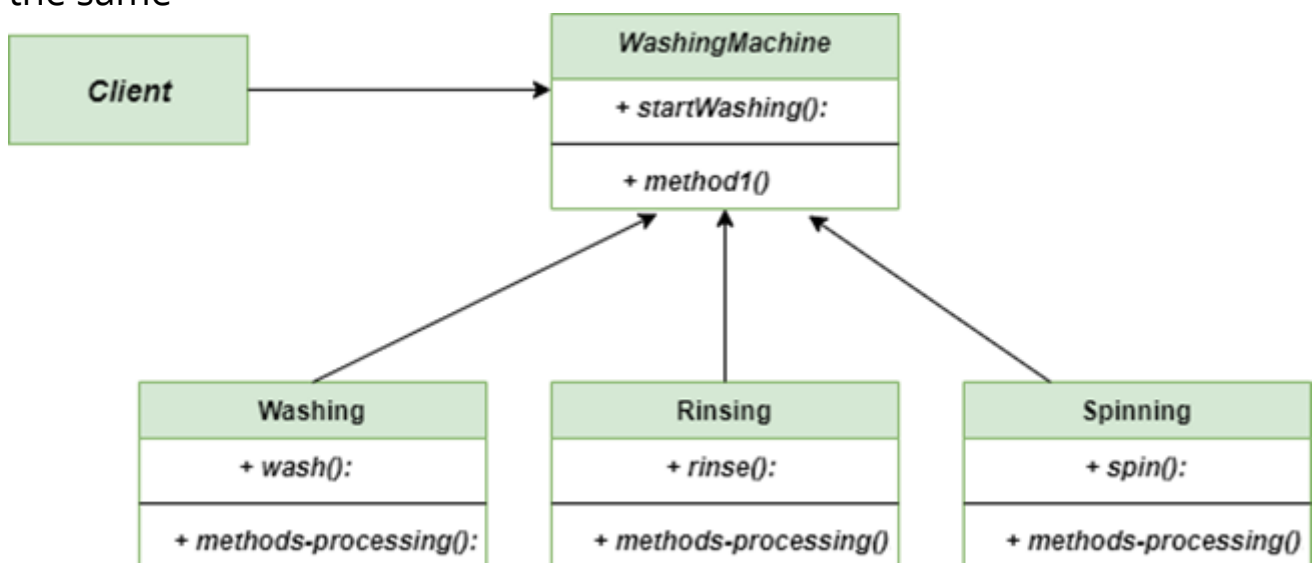
4.a. State the difference between Architectural and Design patterns  
(6.0 Marks)

4.b. Implement the following use case using a proxy pattern. Write both the class diagram and the code snippet. State which type of proxy pattern is suitable for the given use case.

Use Case: If an entire library of books, with all the details about each book, is loaded from a database, it will consume a lot of RAM, and it is very likely that the user will need to issue or return only one book. The solution, that uses the proxy design pattern, displays only the name, author, and availability of the books when a list of them is displayed. When a book is selected, all of the remaining details will be fetched from the database and operations like issue/return (update availability) can be performed.

Note: write only code snippets/pseudo code  
(6.0 Marks)

4.c. Which design pattern is depicted in the given diagram below and describe the pattern in a line or two. Give the pseudo code to implement the same



(8.0 Marks)

5.a. When to use Behavioral patterns? Describe in a line or two interpreter and iterator pattern (4.0 Marks)

5.b. What is an Anti pattern? Briefly describe vendor lock-in anti pattern. (6.0 Marks)

5.c. For the given scenario identify which behavioral pattern can be used? Briefly describe the chosen pattern and draw the class diagram for the scenario with the chosen pattern. (2+3+5)

**Scenario**

An enterprise has been getting more email than they can handle. The enterprise gets 4 types of e-mail.

They are:

- Fan mail with compliments,
- Complaints,
- Requests for new features,
- Spam.

Your task:

- The enterprise has already written the AI detectors that can tell whether an email is fan, complaint, request or spam,
- You need to create a design that can use the detectors to handle incoming email.

(10.0 Marks)



**END SEMESTER ASSESSMENT (ESA) - Jan - May 2024**

**UE19CS353 - Object Oriented Analysis and Design with Java**

**Total Marks : 100.0**

1.a. Briefly describe the need of garbage collector in java.

(4.0 Marks)

1.b. **What is the need for constructors in Java? With an example java program explain default and parameterized constructors** (6.0 Marks)

1.c. **Briefly describe Abstraction, Inheritance and Polymorphism concepts of Object oriented programming** (6.0 Marks)

**1.d. Explain any two different types of inheritance in java with an example for each.** (4.0 Marks)

**2.a. Describe in a line or two the following elements of component diagram: Port, Dependency, Component, Interface.** (4.0 Marks)

**2.b. Draw a use case diagram for ATM machine** (6.0 Marks)

**2.c. What are Generic classes in java. Write an example program to demonstrate generic class in java.** (6.0 Marks)

2.d. Explain in a line or two the following terms with respect to a class diagram

Association

Aggregation

Composition

Qualified Association

(4.0 Marks)

3.a. Draw an activity diagram to login and logout for a web based application

(6.0 Marks)

3.b. With respect to state diagram describe the following terms in line or two and give an example for each.

State

Event

Guard condition

(6.0 Marks)

3.c. Explain in a line or two the following GRASP principles with an example.

i) Creator

ii) Low Coupling

iii) Controller

iv) Polymorphism

(8.0 Marks)



4.a. Describe any four solid design principles with an example for each. (8.0 Marks)

4.b. What is a singleton class. How to make a class singleton (6.0 Marks)

4.c. What are Design patterns? Which are the different types of design patterns, describe in a line or two? Specify pros and cons of design pattern (6.0 Marks)

5.a. Give the UML diagram of the Adapter pattern and briefly describe each component (8.0 Marks)

5.b. Briefly describe the intent, structure and participants of chain of responsibility design pattern. (7.0 Marks)

5.c. What is an Anti pattern? Briefly describe Lava-Flow anti pattern (5.0 Marks)

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**END SEMESTER ASSESSMENT (ESA) - May 2023****UE19CS353 - Object Oriented Analysis and Design with Java****Total Marks : 100.0**

1.a. Elucidate the four main pillars Abstraction, Encapsulation, Inheritance and Polymorphism of Object Oriented Programming in two to three lines. (8.0 Marks)

1.b. State the reason why the given code prints output as false. Also change the code such that the output in the main function is true.

```
public class Sample {  
    public static void main(String[] args) {  
        Demo objOne=new Demo(10,20);  
        Demo objTwo=new Demo(10,20);  
        System.out.println(objOne.equals(objTwo)); //prints false  
    }  
}
```

```
class Demo{  
    int a;  
    int b;  
    Demo(int a,int b){  
        this.a=a;  
        this.b=b;  
    }  
}
```

**(6.0 Marks)**

1.c. For the following code snippet,

1. Output expected from Line 1 of the code is "SRN of the student is 10 and name is pranitha". Add the required functionalities to the class student to get the desired output.

2. Predict the output of Line 2

```
class student
{
int srn;
String name;
// add required functions
}
class Demo
{
public static void main(String args[])
{
student s1 = new student(10,"pranitha");
System.out.println(s1); // Line 1
Student s2=new student(11,"abhay");
System.out.println(s1==s2); // line 2
}
}
```

(6.0 Marks)

2.a. Identity the relationship (include/extend/generalization) between these pair of use cases and depict the relationship between them using appropriate notation.

- 1: Create new order and validate customer account
- 2: Update order and validate customer account
- 3: Place order and login
- 4: Login account and change password
- 5: Choose folder and upload document
- 6: Purchase item, purchase phone, purchase accessories

(6.0 Marks)

2.b. **Draw a Use Case diagram for a Movie Ticket Booking System considering the scenario below:**

Customer can login to the system to utilise its features. System has provision to check currently streaming movies and upcoming movies. The customer can book a ticket and optionally choose a seat of their choice and accordingly make a card payment for it. The bank validates the payment for the system. Customers can also cancel their ticket. The system admin processes the cancellation as well as update movies when required. Customers can provide their review about a movie.

(8.0 Marks)

2.c. Elucidate qualified association and aggregation relationship w.r.t class diagram with an example for each. (6.0 Marks)

**3.a. Draw a State diagram for the book object based on the following scenario:**

Librarians categorise the library books into loanable and non-loanable books. The non-loanable books are the reference books. However, the loanable books are the non-reference books. After cataloguing the books, the books are available for loan. Students who borrow the library books should return them back before the due date. Books that are 12 months over the due date would be considered as a lost state. However, if those books are found in the future, they must be returned back to the library. When the books are found not required in the library or have been damaged, the book would be disposed (8.0 Marks)

**3.b. Write an activity diagram for Dining Philosophers problem:** Five silent philosophers sit at a round table with bowls of noodles. A chopstick is placed between each pair of adjacent philosophers. Each philosopher must alternately think and eat. However, a philosopher can only eat noodles when he has both "left" and "right" chopsticks. Each chopstick can be held by only one philosopher and so a philosopher can use the chopstick only if it is not being used by another philosopher. After he finishes eating, he needs to put down both chopsticks so they become available to others. A philosopher can take the chopstick on his right or the one on his left as they become available, but cannot start eating before getting both of them, Nor can he take a chopstick that is not immediately on his left or right. The problem is how to design a discipline of behaviour such that no philosopher will starve (6.0 Marks)

3.c. With an appropriate example, explain **controller** and **high cohesion** principles in GRASP (6.0 Marks)

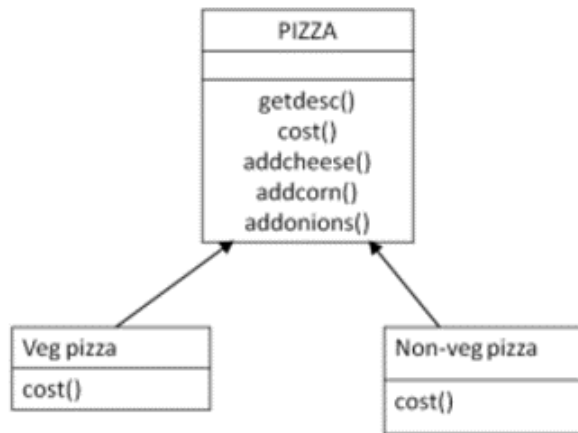
4.a. With an example show how a class can be made a singleton and what are the benefits of making a class a singleton. (5.0 Marks)

4.b. Match the following Design Principle names with their definition.

|                        |                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1: Information Expert  | a. Combine related responsibilities into one manageable unit.                                                               |
| 2. High Cohesion       | b. Provide a well defined interface so that there                                                                           |
| 3. Indirection         | c. Assign responsibility to the class which has the required data to fulfill that responsibility.                           |
| 4. Protected Variation | d. Introduce an intermediate unit to communicate between the other units, so that the other units are not directly coupled. |

(4.0 Marks)

4.c. A software designer for a pizza store has proposed design as shown below.



Identify the issues in proposed design and suggest an appropriate class diagram to address the issues. (7.0 Marks)

4.d. Fill in the blanks:

- i. Violating Dependency Inversion Principle will result in a code with \_\_\_\_\_ coupling.
- ii. Liskov Substitution Principle states that the designer/implementer should be able to replace objects of \_\_\_\_\_ with the objects of \_\_\_\_\_.
- iii. \_\_\_\_\_ Principle states that the design should be extensible however modifying existing design should be avoided.
- iv. The pattern that hides the complexity related to creation of objects from the client is \_\_\_\_\_.

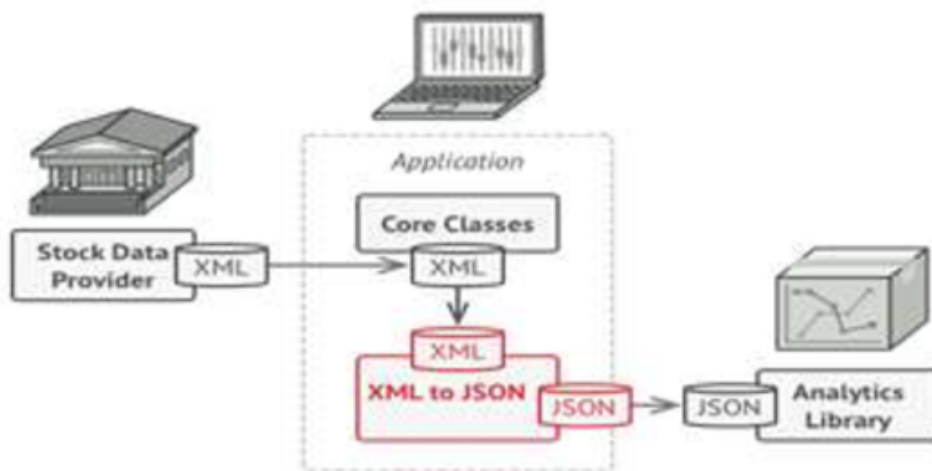
(4.0 Marks)

5.a. Describe the Chain of responsibilities pattern with an example. (4.0 Marks)

5.b. A System has a MediaPlayer interface and a concrete class AudioPlayer implementing the MediaPlayer interface. AudioPlayer can play mp3 format audio files by default. We are having another interface AdvancedMediaPlayer and concrete classes implementing the AdvancedMediaPlayer interface. These classes can play vlc and mp4 format files. We want to make AudioPlayer to play other formats as well.

Identify which design pattern is required to implement the given scenario and Justify. Draw an appropriate class diagram for the scenario with the design pattern.  
(6.0 Marks)

5.c. For a case study depicted in the diagram shown below, identify the appropriate structural design pattern. Write the class structure of the identified pattern.  
(6.0 Marks)



5.d. What is an antipattern? State the solution for Blob and Vendor lock-in antipattern.  
(4.0 Marks)





**END SEMESTER ASSESSMENT (ESA) - JULY - 2023**

**UE19CS353 - Object Oriented Analysis and Design with Java**

**Total Marks : 100.0**

1.a. State any six differences between abstract class and interface? (6.0 Marks)

1.b. Which are the different types of constructors? Give an example to illustrate creation of objects using different types of constructors. (4.0 Marks)

1.c. i) Give an example for Method overloading and Method overriding.  
ii) State the differences between method overloading and method overriding. (6.0 Marks)

1.d. Briefly describe the need of garbage collector in java

(4.0 Marks)

2.a. Elucidate qualified association and aggregation relationship w.r.t class diagram with an example for each.

(6.0 Marks)

2.b. Draw a use case diagram for railway reservation system which has all basic features similar to IRCTC portal.

(6.0 Marks)

2.c. Describe in a line or two the following elements of component diagram: Port, Dependency, Component, Interface.

(4.0 Marks)

2.d. Differentiate between Aggregation and Composition relationships wrt class diagram (4.0 Marks)

3.a. For the given scenarios, choose appropriate diagram to model the scenario. Justify your choice and draw the chosen diagram.

- i) Every student is assigned a Faculty Advisor. The Faculty Advisor is assigned a minimum of 25 students with no maximum limit. Deans and Chairpersons are not assigned any students as advisees. Typically, about half the strength of a section is assigned to a Faculty advisor, i.e., a section typically has two Faculty Advisors.
- ii) An Order consists of multiple Items. The Order total is computed by adding the product of Item Quantity and Item price for each Item. (6.0 Marks)

3.b. Draw an activity diagram for the given below scenario of purchase ticket: Activity is started by Commuter actor who needs to buy a ticket. Ticket vending machine will request trip information from Commuter. This information will include number and type of tickets, e.g. whether it is a monthly pass, one way or round ticket, route number, destination or zone number, etc. Based on the provided trip info ticket vending machine will calculate payment due and request payment options. Those options include payment by cash, or by credit or debit card. If payment by card was selected by Commuter, another actor, Bank will participate in the activity by authorizing the payment (6.0 Marks)

3.c. Explain the creator and controller GRASP principles with an example along with the corresponding class diagram (8.0 Marks)

4.a. With an example show how a class can be made singleton and what are the benefits of making a class a singleton (4.0 Marks)

4.b. What is a Design pattern? Describe in a line or two the essential elements of a design pattern (6.0 Marks)

4.c. i) Specify when to use a builder pattern  
ii) State the difference between Factory and Abstract Factory patter. (4.0 Marks)

4.d. Describe briefly the following:  
1. Liskov Substituiton Principle  
2. Interface segregation principle  
3. Open-Close principle (6.0 Marks)

5.a. A System has a MediaPlayer interface and a concrete class AudioPlayer implementing the MediaPlayer interface. AudioPlayer can play mp3 format audio files by default. We are having another interface AdvancedMediaPlayer and concrete classes implementing the AdvancedMediaPlayer interface. These classes can play vlc and mp4 format files. We want to make AudioPlayer to play other formats as well.

Identify which design pattern is required to implement the given scenario and Justify. Draw an appropriate class diagram for the scenario with the design pattern.  
(6.0 Marks)

5.b. Differentiate between pattern and anti-pattern.

(4.0 Marks)

5.c. What is a proxy pattern? Which are the three main variations of proxy pattern.  
(6.0 Marks)

5.d. State the solution for Blob, Lava Flow, Vendor lockin and Analysis-Paralysis antipatterns  
(4.0 Marks)



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**END SEMESTER ASSESSMENT (ESA) - DEC 2023**

**UE19CS353 - Object Oriented Analysis and Design with Java**

**Total Marks : 100.0**

1.a. Which are the three types of constructors that can be defined in Java?  
Give examples for each. (6.0 Marks)

1.b. What is inheritance, explain different types of inheritance in java with  
an example for each. (6.0 Marks)

1.c. Briefly describe the need for garbage collector in java. (4.0 Marks)

1.d. Write the output of the following java program

```
class A {  
    void show() {  
        System.out.println("show of class A");  
    }  
}  
class B extends A {  
    void show() {  
        System.out.println("show of class B");  
    }  
  
    void disp() {  
        System.out.println("show of class B disp");  
    }  
}  
class supertest1 {  
    public static void main(String args[]) {  
        A aobj = new A();  
        B bobj = new B();  
        A refa;  
        refa = aobj;  
        refa.show();  
        refa = bobj;  
        refa.show();  
        if (refa instanceof B) {  
            ((B) refa).disp();  
        }  
    }  
}
```

(4.0 Marks)

2.a. Elucidate qualified association and aggregation relationship w.r.t class diagram with an example for each (6.0 Marks)

2.b. Describe in a line or two the following elements of component diagram: Port, Dependency, Component, Interface (4.0 Marks)

2.c. Draw an use case diagram for e-commerce website (6.0 Marks)

2.d. What are Generic classes in java. Write an example program to demonstrate any one generic class in java. (4.0 Marks)



3.a. i) Match the following Design Principle names with their definition. 4M

|                        |                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1. Information Expert  | a. Combine related responsibilities into one manageable unit.                                                               |
| 2. High Cohesion       | b. Provide a well defined interface so that there will be no affect on other units                                          |
| 3. Indirection         | c. Assign responsibility to the class which has the required data to fulfill that responsibility                            |
| 4. Protected Variation | d. Introduce an intermediate unit to communicate between the other units, so that the other units are not directly coupled. |

ii) Differentiate between guard condition and change event in state modeling. 2M (6.0 Marks)

3.b. Explain in a line or two the following GRASP principles with an example.

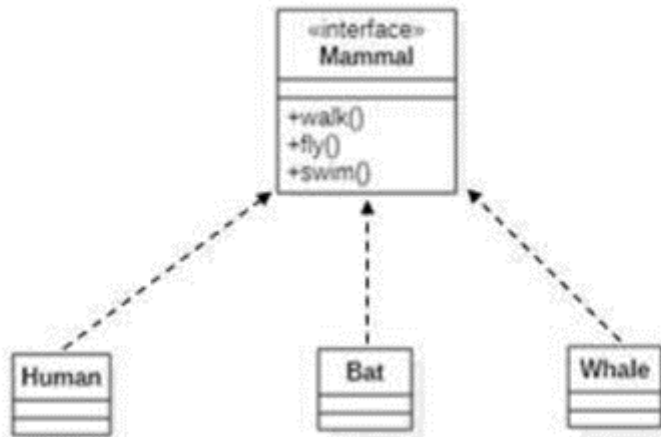
- i) Creator
- ii) Low Coupling
- iii) Controller
- iv) Polymorphism

(8.0 Marks)

3.c. Draw an activity diagram to perform withdrawal operation in an ATM machine. Specify all the activities and events along with swimlanes. (6.0 Marks)

4.a. What SOLID design principle is violated in the given design? How can this be modified to make it a better design? Draw class diagram and code snippet of your design.

(8.0 Marks)

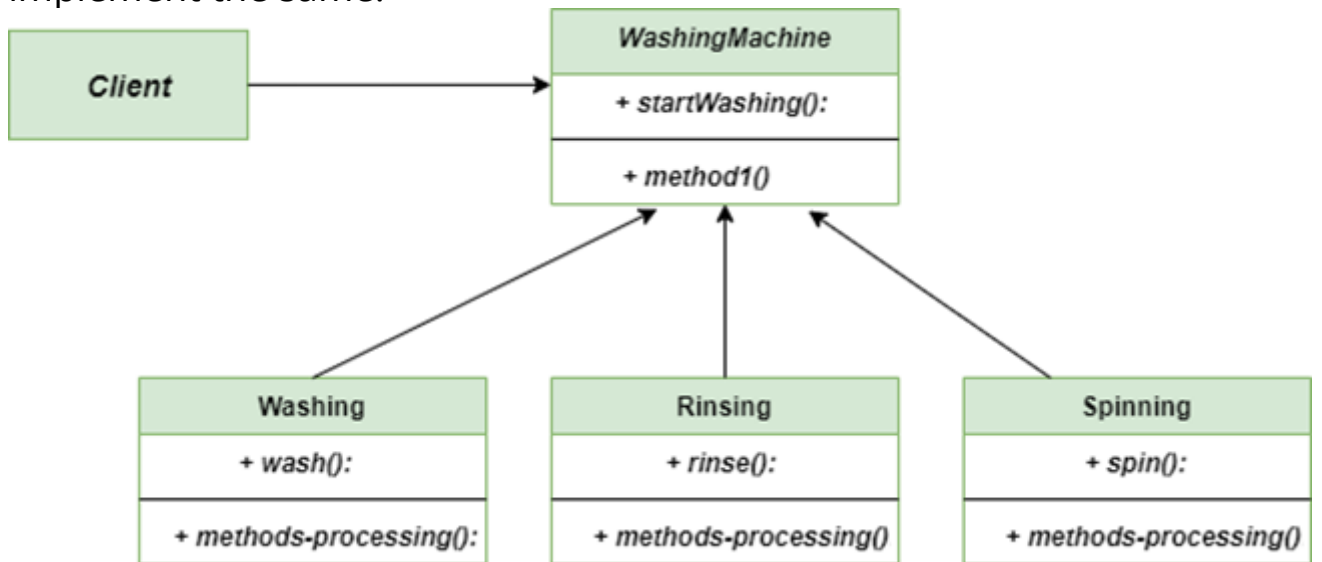


4.b. Describe the Singleton creational pattern with a suitable code snippet (6.0 Marks)

4.c. What are Design patterns? Which are the different types of design patterns, describe in a line or two? Specify pros and cons of design pattern. (2+2+2) (6.0 Marks)

5.a. Give the UML diagram of the Adapter pattern and briefly describe each component (7.0 Marks)

5.b. Which design pattern is depicted in the given diagram below? Describe the identified pattern in a line or two. Give the pseudo code to implement the same.



(8.0 Marks)

5.c. What is an Anti pattern? Briefly describe vendor lock-in anti pattern (5.0 Marks)

**END SEMESTER ASSESSMENT (ESA) - Jan - May 2024****UE18CS353 - Object Oriented Analysis and Design with Software Engineering****Total Marks : 100.0**

1.a. Using a skeletal system block diagram, compare and contrast Iterative and Incremental Development Models. (6.0 Marks)

1.b. What is the problem with the following requirement statement? That is, what quality attributes the statement does not possess? How can it be improved?

**“SR-4: All pages of the application will load in acceptable amount of time.”**  
(6.0 Marks)

1.c. Describe what is a risk and the steps (with 1 line description) involved in managing risks. (8.0 Marks)

2.a. Describe briefly (1-2 lines) any six approaches that can be taken to decompose a large system into subsystems or components.

(6.0 Marks)

2.b. Draw the Class Diagram to model a system for management of courses and enrolments. Identify classes, attributes, relationships and cardinality.

1. A Department manages multiple Sessions.
2. A Session has a Start date, an End date and Name (like Jan-May 2023).
3. A session can have multiple semesters running (like 6<sup>th</sup>, 4<sup>th</sup>, 2<sup>nd</sup> etc).
4. A student belongs to a semester and to a section within it. A section has minimum of 60 students.
5. A semester has a set of courses offered in it.
6. A student has to enrol for all the required number of courses (let us assume 5 for the listed semesters). For some courses like electives, a minimum of 20 students have to enrol for it to be offered.
7. An enrolled student's attendance for the course is recorded.
8. Similarly, an enrolled student's ISA, ESA marks and Grade for the course are recorded.
9. A teacher will be assigned to a section for a course.
10. A teacher can also be assigned as the class teacher for a particular section

Note: Assume attributes and other details as appropriate where not provided.

(8.0 Marks)

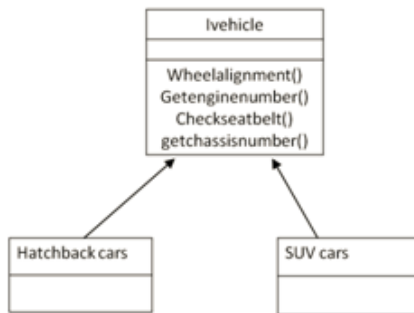
2.c. For the given scenarios, choose appropriate diagram to model the scenario. Justify your choice and draw the chosen diagram.

- i) Every student is assigned a Faculty Advisor. The Faculty Advisor is assigned a minimum of 25 students with no maximum limit. Deans and Chairpersons are not assigned any students as advisees. Typically, about half the strength of a section is assigned to a Faculty advisor, i.e., a section typically has two Faculty Advisors.
- ii) An Order consists of multiple Items. The Order total is computed by adding the product of Item Quantity and Item price for each Item.

(6.0 Marks)

3.a. Class design given below has functionalities of vehicles. Now, the client wants to add a two wheeler class to the design. If the new two wheeler class implements interface lvehicle, which design principle is violated? Modify the design in the given class diagram to accommodate the new class.

(8.0 Marks)



3.b. Differentiate between Coding standards and guidelines for an organization with at least 2 examples for each.

(6.0 Marks)

3.c. What is secure programming? List the secure programming approaches or practices to overcome software vulnerabilities.

(6.0 Marks)

4.a. Choose most appropriate types of testing applicable in the following applications from the list of testing types given below:

**Functional, System, Load, Stress, Performance, Compatibility, Security, Reliability, Availability, Regression, Usability, Acceptance**

1. Online shopping website (like Amazon), where they offer special sale days (like Prime Days).
2. A file management system (like Google Drive), where different types of files can be uploaded and viewed from multiple devices. It also needs to provide collaborative editing feature.
3. Anti-Virus software that needs to be kept up-to-date (in terms of software and data related to newer vulnerabilities) frequently. (6.0 Marks)

4.b. What are the 3 perspectives for looking at test related measures? Give one example for each. (6.0 Marks)

4.c. What is SEI CMM? Name two reasons why you would want to use it? Enumerate its five levels. (8.0 Marks)

5.a. List any six objectives of IT Systems Management (ITSM). (6.0 Marks)

5.b. Name any three challenges of Geographical distance with respect to Software Development? How can they be overcome? (6.0 Marks)

5.c. Describe the term DevOps. Write briefly about the DevOps pipe line. (8.0 Marks)



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**END SEMESTER ASSESSMENT (ESA) - May 2023****UE18CS353 - Object Oriented Analysis and Design with Software Engineering****Total Marks : 100.0**

1.a. Discuss different phases in the development of software. (5.0 Marks)

1.b. Illustrate the advantages and disadvantages of the Agile SCRUM model with an example. (5.0 Marks)

1.c. Differentiate the Incremental model and Evolutionary Model (5.0 Marks)

1.d. How will a software development project is planned using SE Management?  
(5.0 Marks)

2.a. Consider a library management system. Draw a use case diagram that represents the primary functionalities of the system. Include at least three actors and five use cases. Provide appropriate labels and relationships between the actors and use cases. Briefly explain the significance of one of the relationships in the diagram.  
(5.0 Marks)

2.b. Explain the various types of relationships that can be depicted between classes in a class diagram, and provide an illustrative example to support your explanation.  
(5.0 Marks)

2.c. Consider a process for order fulfilment in an e-commerce system. Create an activity diagram that includes swimlanes, and demonstrates the use of fork and join constructs. Include at least three swimlanes representing different roles or departments involved in the process. Clearly depict the actions within each swimlane, and illustrate the parallelism and synchronization of activities using the fork and join constructs. Briefly explain the significance of using swimlanes, fork, and join in this activity diagram.  
(5.0 Marks)

2.d. Elaborate on any one architectural design pattern with illustrative example  
(5.0 Marks)

3.a. How does the Creator pattern in GRASP help in assigning responsibilities?  
(5.0 Marks)

3.b. Describe the Liskov Substitution Principle (LSP) and its implications for object-oriented design.  
(5.0 Marks)

3.c.

i. Match the following Design Principle names with their definition.

|                        |                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1. Information Expert  | a. Combine related responsibilities into one manageable unit.                                                               |
| 2. High Cohesion       | b. Provide a well defined interface so that there will be no affect on other units                                          |
| 3. Indirection         | c. Assign responsibility to the class which has the required data to fulfill that responsibility.                           |
| 4. Protected Variation | d. Introduce an intermediate unit to communicate between the other units, so that the other units are not directly coupled. |

ii. State True or False with justification:

- Protected Variation can be achieved using Polymorphism.
- Pure Fabrication classes cannot be reused in other systems.

(6.0 Marks)

3.d. Describe the role of version control in a Configuration Management System.  
(4.0 Marks)

4.a. Describe the V-model of the software life cycle and discuss how testing activities are integrated into each stage.  
(5.0 Marks)

4.b. Compare and contrast verification and validation in software testing. Provide examples to illustrate each concept.  
(5.0 Marks)

4.c. Describe statement coverage and decision coverage as coverage-based test techniques. Discuss the differences between these two techniques and their importance in ensuring comprehensive test coverage. (5.0 Marks)

4.d. Explain the principles behind mutation testing and fault injection. Discuss their advantages and limitations as fault-based test techniques. (5.0 Marks)

5.a. Describe the ITIL service lifecycle and its five stages. Discuss the purpose and activities involved in each stage. (5.0 Marks)

5.b. Discuss the concept of affinity in DevOps and how it fosters cross-functional cooperation and shared responsibilities. Provide examples of how affinity can be established in a DevOps environment. (5.0 Marks)

5.c. Explain how continuous build, integration, and delivery contribute to shorter development cycles and faster time-to-market. Discuss the role of tools like Jenkins in facilitating these practices. (5.0 Marks)

5.d. Discuss the importance of automated build, testing, and deployment in the DevOps workflow. Explain how automation helps reduce errors and improve the reliability of software releases. (5.0 Marks)

**END SEMESTER ASSESSMENT (ESA) - JULY - 2023****UE18CS353 - Object Oriented Analysis and Design with Software Engineering****Total Marks : 100.0**

1.a. Briefly explain about Requirements Engineering Process with a neat diagram

(6.0 Marks)

1.b. Write a use case diagram for Online Shopping website, considering any e-commerce application.

(6.0 Marks)

1.c. Discuss the V Model of Software Development Lifecycle with the different phases, focus of each of these phases and the deliverables or artifacts which you would see at the end of the phase. Discuss how is this similar to a waterfall model and how is it different from a waterfall model.

(8.0 Marks)

2.a. Contrast Architectural Views, Styles and Patterns. Give an example for each

(6.0 Marks)

2.b. Define the following wrt the class diagram:

- a) Association
- b) Aggregation
- c) Composition
- d) Multiplicity
- e) Qualifier
- f) Generalization

(6.0 Marks)

2.c. Develop an activity diagram for login of an application. Identify atleast 4 activities, specify all the transitions. (8.0 Marks)

3.a. Explain SOLID design principles. Give an example for any 2 principles (8.0 Marks)

3.b. Define the following: Preventive Maintenance, Perfective Maintenance, Corrective Maintenance, Adaptive Maintenance (4.0 Marks)

3.c. Define the following wrt Quality of an Agile Scrum project:

- i) Sprint Burndown
- ii) Team Velocity Metric
- iii) Throughput
- iv) Cycle Time

(4.0 Marks)

3.d. Enlist the benefits of using Software configuration System. (4.0 Marks)

4.a. Differentiate between Static Testing and Dynamic Testing (6.0 Marks)



|                                                                                                     |             |
|-----------------------------------------------------------------------------------------------------|-------------|
| 4.b. Contrast between Manual and Automation testing                                                 | (6.0 Marks) |
| 4.c. Briefly discuss about different levels of testing.                                             | (8.0 Marks) |
| 5.a.<br>Discuss the following:<br>1: Hacking and Software engineering<br>2: Availability Management | (6.0 Marks) |
| 5.b. Enlist the pillars of Devops and specify few factors about each pillar of Devops               | (8.0 Marks) |
| 5.c. Briefly discuss about ITSM and ITIL                                                            | (6.0 Marks) |



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**END SEMESTER ASSESSMENT (ESA) - DEC 2023**

**UE18CS353 - Object Oriented Analysis and Design with Software Engineering**

**Total Marks : 100.0**

1.a. With Diagram, Explain the software development life cycle(SDLC)  
(8.0 Marks)

1.b. List 8 activities that are performed during the feasibility study of the requirement analysis.  
(8.0 Marks)

1.c. Explain the advantages of waterfall model  
(4.0 Marks)

2.a. Identify the at least 6 classes from the below case study:

Design the software to support a computerized banking network including both human cashiers and automatic teller machines (ATMs) to be shared by a consortium of banks. Each bank provides its computer to maintain its own accounts and process transactions against them. Cashier stations are owned by individual banks and communicate directly with their own bank's computers. Human cashiers enter account and transaction data. Automatic teller machines communicate with a central computer that clears transactions with the appropriate banks. An automatic teller machine accepts a cash card, interacts with the user, communicates with the central system to carry out the transaction, dispenses cash, and prints receipts. The system requires appropriate recordkeeping and security provisions. The system must handle concurrent access to the same account correctly.

The banks will provide their software for their computers; you are to design the software for the ATMs and the network. The cost of the shared system will be apportioned to the banks according to the number of customers with cash cards. (6.0 Marks)

2.b. Draw Component diagram for ATM Components (8.0 Marks)

2.c. Draw a sequence diagram for a phone calling application (6.0 Marks)

3.a. Explain the single responsibility principle

(6.0 Marks)

3.b. Explain briefly the code review and code inspection

(8.0 Marks)

3.c. Explain Software Configuration Management and its benefits

(6.0 Marks)

4.a. Explain briefly about the 4 pillars of DevOps

(8.0 Marks)

4.b. With Diagram explain the levels of testing

(8.0 Marks)

4.c. Write a note on Black box & White box testing

(4.0 Marks)

5.a. Explain the key issues in the software maintenance

(8.0 Marks)

5.b. With Diagram explain the process of test execution

(6.0 Marks)

5.c. Differentiate between VERIFICATION & VALIDATION

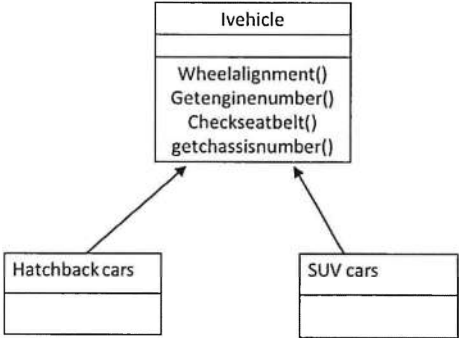
(6.0 Marks)



**JULY 2021: END SEMESTER ASSESSMENT (ESA) B TECH VI SEMESTER**

**UE18CS353 – OBJECT ORIENTED ANALYSIS AND DESIGN WITH  
SOFTWARE ENGINEERING**

| Time: 3 Hrs |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Answer All Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Max Marks: 100 |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1           | a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | What is a software life cycle model? What are the common characteristics of each phase or activity in a software life cycle model?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5              |
|             | b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | What is the problem with the following requirement statement? That is, what quality attributes the statement does not possess? How can it be improved?<br>"SR-4: The pages of the application will load in acceptable amount of time."                                                                                                                                                                                                                                                                                                                                                                                                        | 4              |
|             | c)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Answer the following:<br>1. The model where the existing code is not modified to a great extent while adding new features is _____. The model where the existing code is modified to a great extent while adding new features is _____.<br>2. While preparing the project plan, we need to plan for documentation activities as well. True or False.<br>3. In Scrum model, at the end of every week, a working version of the software should be available. True or False.<br>4. An advantage of Component based software engineering is that Components could be used as a black box and hence reduces complexity of systems. True or False. | 5              |
|             | d)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Discuss what is a risk and the steps (with 1 line description) involved in managing risks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6              |
| 2           | a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Describe briefly (1-2 lines) any six approaches that can be taken to decompose a larger system into subsystems or components.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6              |
|             | <p align="center"><b>Case Study for Questions 2(b) and 2(c)</b></p> <p>An OTT video service website, CutFlix, needs to have the following features.</p> <ul style="list-style-type: none"> <li>- There are two types of videos, a movie or an episode of a series. The movie has details like name, release year, producer, description, director and duration.</li> <li>- The movie also has associated actors. The platform should display the list of actors and the roles played by them, which may include multiple roles played by the same actor.</li> <li>- An episode belongs to the season of a series. It has a name, release date, producer, director, duration and description.</li> <li>- The season will have details like number of episodes, release year and description. The series has name, description, genres and maturity rating.</li> <li>- Each episode also has associated actors. The platform should display the list of actors and the roles played by them, which may include multiple roles played by the same actor.</li> <li>- A registered user can create multiple profiles. A profile has a list of videos viewed. On deleting the user account, all associated profiles should also be deleted.</li> <li>- A user can add the video to their profile's Watchlist. S/He could remove the video from Watchlist anytime.</li> <li>- The video currently being watched by the user appears in the profile's "Continue watching" list. The user can remove the video from this list at any point of time. The video is automatically removed from the list once the user completes watching the video.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                |
|             | b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | For the above case study, prepare the class diagram by identifying the classes, their properties, associations, association classes and cardinality.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 8              |

|   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
|---|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | c) | For the video object described above, prepare the state diagram identifying the states, transitions, events and activities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6 |
| 3 | a) | <p>Class design given below has functionalities of vehicles. Now, the client wants to add a two wheeler class to the design. If the new two wheeler class implements interface Ivehicle, which design principle is violated? Modify the design in the given class diagram to accommodate the new class.</p>  <pre> classDiagram     class Ivehicle {         &lt;&lt;interface&gt;&gt;         +Wheelalignment()         +Getenginenumber()         +Checkseatbelt()         +getchassisnumber()     }     class Hatchback_cars     class SUV_cars     Hatchback_cars -- &gt; Ivehicle     SUV_cars -- &gt; Ivehicle         </pre>                                                              | 8 |
|   | b) | Differentiate between Coding standards and guidelines for an organization with at least 2 examples for each.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6 |
|   | c) | How do you enable software to be testable? Write briefly about any six of these techniques.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 6 |
| 4 | a) | <p>Choose most appropriate types of testing applicable in the following applications from the list of testing types given below:</p> <p><b>Functional, System, Load, Stress, Performance, Compatibility, Security, Reliability, Availability, Regression, Usability, Acceptance</b></p> <ol style="list-style-type: none"> <li>Online shopping website (like Amazon), where they offer special sale days (like Prime Days).</li> <li>A file management system (like Google Drive), where different types of files can be uploaded and viewed from multiple devices. It also needs to provide collaborative editing feature.</li> <li>Anti-Virus software that needs to be kept up-to-date (in terms of software and data related to newer vulnerabilities) frequently.</li> </ol> | 6 |
|   | b) | <p>Answer and indicate whether these statements are true or false.</p> <ol style="list-style-type: none"> <li>Fault based testing is a Code based testing where a fault is purposely introduced into the code</li> <li>Acceptance Testing is performed anytime during Agile development on request of the customer</li> <li>It is easier to execute Regression testing manually</li> <li>Destructive model is a testing model chosen as part of testing strategy</li> <li>Test adequacy criteria helps to determine when to stop testing</li> <li>Bugs/kLoC is a valid test metric</li> </ol>                                                                                                                                                                                     | 6 |
|   | c) | Contrast the four different types of Software Maintenance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|   | d) | What is SEI CMM? Name two reasons why you would want to use it? Enumerate its five levels.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |



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|   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
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| 5 | a) | Answer the following:<br>1. Some of the most exemplary DevOps initiatives were started in giant and mature IT organizations. True or False.<br>2. An important ethic for a software engineer is to keep in mind the overall good the software will be for the larger society. True or False.<br>3. Trying out solutions until an optimal solution is created is _____ and working with one solution that suits all constraints and requirements is _____. | 4 |
|   | b) | What are the challenges of Geographical distance with respect to Software Development? How can they be overcome?                                                                                                                                                                                                                                                                                                                                          | 8 |
|   | c) | Describe the term DevOps. Write briefly about the DevOps pipe line.                                                                                                                                                                                                                                                                                                                                                                                       | 8 |

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**Seventh Semester B.E. Degree Examination, June-July 2009**  
**Object Oriented Analysis and Design**

Time: 3 hrs.

Max. Marks:100

**Note: Answer any FIVE full questions.**

- 1 a. What are the main advantages of object oriented development? Explain. (08 Marks)  
 b. With suitable examples explain following:  
    i) Object and Identity. (08 Marks)  
    ii) Static and dynamics binding of objects. (08 Marks)  
 c. List and explain Coad and Yourdon clues for finding candidate classes and objects. (04 Marks)
- 2 a. How to achieve high quality in software? List and explain four quality measures for software evaluation. (08 Marks)  
 b. What is the importance of prototyping? What are the commonly accepted prototyping? What are their strengths? (06 Marks)  
 c. What is the specific aim of OOSE? What is an objectory? What are the different models with which the objectory is built? (06 Marks)
- 3 a. Describe Object Modeling Technique (OMT) by Jim Kumbaregh Et.Al. (08 Marks)  
 b. Define patterns and frame works. List major differences between patterns and frame works. (06 Marks)  
 c. What are the significances of Antipatterns and Generative patterns? (06 Marks)
- 4 a. What are the advantages of modeling? What elements a modeling language must include? (06 Marks)  
 b. Write the UML sequence and collaboration diagrams for the telephone call. (08 Marks)  
 c. Draw the UML state chart diagram to show different states and events responsible for the change of states of a process cooperating systems. (06 Marks)
- 5 a. What are the advantages of developing a business process model? With an example of library members activities, explain how an activity diagram aids in developing use-case. (10 Marks)  
 b. What are the guidelines for developing effective documentation? Explain. (05 Marks)  
 c. What is object classification? What are the different approaches for identifying classes? Explain each briefly. (05 Marks)
- 6 a. How the classes are identified using common class patterns approach? Explain. (06 Marks)  
 b. For Via Net ATM bank system develop sequence and collaboration diagrams for deposit checking use-case. (06 Marks)  
 c. What is an association? How do you identify associations? Explain how to eliminate unnecessary associations. (08 Marks)
- 7 a. What are coupling and cohesion of objects or software components? (06 Marks)  
 b. Write a note on class visibility. (06 Marks)  
 c. Explain UML attribute presentation. Give complete UML class diagram for via Net Bank system. (08 Marks)
- 8 a. What is OODBMS? Explain. Discuss the differences between OODBMS and traditional databases. (08 Marks)  
 b. Discuss the four major activities of designing view layer classes. (08 Marks)  
 c. What are the guidelines for designing application windows? Discuss. (04 Marks)



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| <b>NEW SCHEME</b> |
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**Seventh Semester B.E. Degree Examination, May 2007**  
**CS / IS**

**Object Oriented Analysis and Design**

Time: 3 hrs.]

[Max. Marks:100

**Note : Answer any FIVE full questions.**

- 1
  - a. What is object-oriented system development methodology? Explain the reasons for using object orientation in system development. (10 Marks)
  - b. With examples, explain the following with reference to object oriented systems:
    - i) Class hierarchy.
    - ii) Polymorphism. (10 Marks)
- 2
  - a. Describe briefly the macro processes involved in the object oriented system development approach. (12 Marks)
  - b. What are the quality measures that need to be considered for building high quality software? (08 Marks)
- 3
  - a. Briefly describe the Booch system development processes. (12 Marks)
  - b. Discuss the importance of Rumbaugh's OMT. Also describe the way in which OMT is modeled. (08 Marks)
- 4
  - a. What is UML? Why is it important? Explain with examples any two UML diagrams. (12 Marks)
  - b. With a block diagram, explain the object oriented analysis process in the unified approach. (08 Marks)
- 5
  - a. Explain the following relationships with examples:
    - i) Super sub class.
    - ii) A-part-of (10 Marks)
  - b. What is association? Explain with an example. Also explain the ways of eliminating unnecessary associations. (10 Marks)
- 6
  - a. List the object oriented design axioms and corollaries. (10 Marks)
  - b. Write briefly on:
    - i) CORBA
    - ii) Multidatabase systems. (10 Marks)
- 7
  - a. Explain the macro level design process and micro level design process of view layer. (10 Marks)
  - b. Explain the object oriented design process in the unified approach. (10 Marks)
- 8
 

Write short notes on:

  - a. Frame works.
  - b. Design patterns.
  - c. Interface design as a creative process.
  - d. Inheritance. (20 Marks)



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**Seventh Semester B.E. Degree Examination, Dec.08/Jan.09**  
**Object Oriented Analysis & Design**

Time: 3 hrs.

Max. Marks:100

**Note: Answer any FIVE full questions.**

- 1
  - a. Why object orientation is necessary? Give reasons. (04 Marks)
  - b. Differentiate between object oriented approach and structured approach. (06 Marks)
  - c. Describe the following terms and give an example for each. (10 Marks)
    - (i) Class hierarchy
    - (ii) Inheritance
    - (iii) Polymorphism
    - (iv) Aggregation
    - (v) Object Identity
- 2
  - a. Using an example of Water-Fall approach, explain how the software development process can be viewed as series of transformation. (06 Marks)
  - b. Describe the quality measures for building high-quality software. (04 Marks)
  - c. With neat diagram, explain the macroprocesses of the object-oriented system development approach. (10 Marks)
- 3
  - a. What are the phases of object modeling technology? Describe each phase briefly. (04 Marks)
  - b. Explain the Booch system development processes. (06 Marks)
  - c. Explain the following methodologies: (i) Use case (ii) OOSE (iii) OOSE (10 Marks)
- 4
  - a. Define pattern. Explain the essential components on reading a pattern. (06 Marks)
  - b. Give the differences between design pattern and frameworks. (04 Marks)
  - c. With neat diagram, explain the processes and components of the Unified approach. (10 Marks)
- 5
  - a. How uses and extends associations are useful in use-case modeling? Describe briefly with an example. (04 Marks)
  - b. Why documentation is an important part of analysis? Explain the guidelines for developing effective documentation. (06 Marks)
  - c. Explain the different use-cases used in Vianet – Bank ATM system. (10 Marks)
- 6
  - a. What are the four main approaches for identifying classes? Explain in detail. (10 Marks)
  - b. Briefly describe the guidelines for identifying super-sub relationship. (04 Marks)
  - c. Define a part-of-relationship with an example. Describe its properties and patterns. (06 Marks)
- 7
  - a. Explain in detail the object oriented design Axioms and Corollaries. (10 Marks)
  - b. Describe briefly the design patterns created by 'Kurotsuchi'. (04 Marks)
  - c. What do you mean by coupling? Explain the different types of coupling by specifying degree of coupling. (06 Marks)
- 8
  - a. Define OODBMS. Give the differences between an OODBMS and traditional DBMS. (04 Marks)
  - b. Explain the characteristics of Multi-database systems. (06 Marks)
  - c. How objects are mapped with relational systems by different mapping capabilities? Explain with an example for each capability. (10 Marks)





**PES Institute of Technology, Bangalore**  
(Autonomous Institute under VTU, Belgaum)

**IS 402****SEMESTER END EXAMINATION (SEE) B. E. 7<sup>th</sup> SEMESTER – DECEMBER 2010****IS 402 – OBJECT ORIENTED ANALYSIS AND DESIGN**

Time: 3 Hrs

Answer All Questions

Max Marks: 100

|    |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |
|----|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1. | a) | Explain briefly the five views for describing the software intensive system architecture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 10 |
|    | b) | Consider a retail store called "XYZ COMPANY". It takes orders from its customers. The customers may be located at different parts of the country. Once the customer orders for product(s), they are checked against the inventory present in their warehouse. If present, an invoice is generated through a set of transactions. Finally, the products are packed and kept ready for delivery through shipment. The company provides free shipment for its customers. Suppose XYZ COMPANY is interested in automating the complete ordering process, identify the classes required along with suitable members and represent them using appropriate UML diagram. | 04 |
|    | c) | i) Mr. James has a CPU. Generally, a CPU may be associated with no or many controllers depending on the usage of disk drives. Mr. James has four disk drives and wants an efficient controller to control these drives. He opts for a special type of controller called SCSI controller which has ability to control a minimum of one disk drive to a maximum of four disk drives. Write a UML diagram to model Mr. James system's components specified in the given scenario.                                                                                                                                                                                   | 02 |
|    |    | ii) Consider the following class Skelton. Analyses and write an equivalent UML diagram for the same.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 02 |
|    |    | <pre> public class Sale {     public void updatePriceFor( ProductDescription description )     {         Money basePrice = description.getPrice();         //...     }     // ...}; </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
|    | d) | What is a stereotype? What are three ways of representing the stereotype?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 02 |
| 2. | a) | i) Write a class diagram for the following object diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 02 |
|    |    | <pre> classDiagram     class Spain {         Name := Spain     }     class France {         Name := France     }     class Belgium {         Name := Belgium     }     Spain -- France : Borders     France -- Belgium : Borders </pre>                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
|    |    | ii) An undirected graph consists of a set of vertices and a set of edges. Edges connect pair of vertices. Prepare a class model to describe such an undirected graph. Note that the model should capture only the structure of the graph i.e. connectivity.                                                                                                                                                                                                                                                                                                                                                                                                      | 03 |
|    | b) | Categorize the following into different types of UML relationships<br>i) CPU and display are parts of computer<br>ii) A user can login as admin or guest.<br>iii) A bank has one or more accounts where each account can be checking, savings or money market.<br><b>Note: Draw suitable class diagrams to illustrate the above relationship. Indicate association and multiplicities.</b>                                                                                                                                                                                                                                                                       | 04 |
|    | c) | Explain briefly the dependency relationships among<br>i. Class diagrams (any five)      ii. Packages      iii. Use cases.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 08 |
|    | d) | The following code illustrates a template class Set fragment that can be used to define behaviour for sets in general.<br><pre> class Set &lt;T&gt; {     public:         void insert (T newElement);         void remove (T anElement);         ..... }; </pre> Write a UML template class to illustrate the model of instantiation of Set <Employee> employeeSet;                                                                                                                                                                                                                                                                                              | 03 |

